Irreducibility in $\mathbb{K}[x]$.

- **Exercise 1**
  Determine polynomials below is rather reducible or not.

$$
\begin{array}{c|c|c|c|c}
 & \mathbb{Z} & \mathbb{Q} & \mathbb{R} & \mathbb{C} \\
p_1(x) = x^2+4x+4 = (x+2)^2 & 1 & 1 & 1 & 1 \\
p_2(x) = x^2-4 = (x-2)(x+2) & 1 & 1 & 1 & 1 \\
p_3(x) = 9x^2-3 = 3\left(\sqrt{3}x-1\right)\left(\sqrt{3}x+1\right) & 0 & 0 & 1 & 1 \\
p_4(x) = x^2-\frac{4}{9} = \left(x-\frac{2}{3}\right)\left(x+\frac{2}{3}\right) & 0 & 1 & 1 & 1 \\
p_5(x) = x^2-2 = \left(x-\sqrt{2}\right)\left(x+\sqrt{2}\right) & 0 & 0 & 1 & 1 \\
p_6(x) = x^2+1 = (x-\mathrm{i})(x+\mathrm{i}) & 0 & 0 & 0 & 1
\end{array}
$$

- **Exercise 2**
  Prove that $x=1$is a root of $p(x)=2x^4-3x^3-3x^2+7x-3$ and find its multiplicity. Is $p(x)$ reducible over $\mathbb{Z},\mathbb{Q},\mathbb{R},\mathbb{C}$?
  - **Solution 2.1**
    $p(1)=2-3-3+7-3=0 \implies x=1$ is a root. Then $p(x)=(x-1)\left(2x^3-x^2-4x+3\right)$. We find that $2-1-4+3=0$. Then $p(x)=(x-1)^2\left(2x^2+x-3\right)$ and then $p(x)=(x-1)^3(2x+3)$. Therefore p is reducible over $\mathbb{Z},\mathbb{Q},\mathbb{R},\mathbb{C}$.

- **Exercise 3**
  FInd a real root of $f(x)=x^3-\sqrt{2}x^2+x-\sqrt{2}$.
  - **Solution 3.1**
    We have that $f(x)=\left(x-\sqrt{2}\right)\left(x^2+1\right)$. Therefore $x=\sqrt{2}$ is a real root.

- **Exercise 4**
  Determine $p(x)=x^{12}+x^6+1$ is reducible or not over $\mathbb{Z},\mathbb{Q},\mathbb{R},\mathbb{C}$.
  - **Solution 4.1**
    We have that

$$
\begin{aligned}
p(x) &= x^{12}+2x^6+1-x^6 \\
&= \left(x^6+1\right)^2-x^6 \\
&= \left(x^6+1-x^3\right)\left(x^6+1+x^3\right).
\end{aligned}
$$

    Therefore it is reducible over $\mathbb{Z},\mathbb{Q},\mathbb{R},\mathbb{C}$.

- **Exercise 5**
  Find the roots of the following polynomial $p(x)=x^5-3x^4+5x^3-x^2-10x$.
  - **Solution 5.1**
    We have that $p(x)=x\left(x^4-3x^3+5x^2-x-10\right)$. By rational roots theorem, possible rational roots are $\pm1,\pm2,\pm5,\pm10$. We find that $p(-1)=p(2)=0$. Thus $p(x)=x(x+1)(x+2)\left(x^2-2x+5\right)$.

- **Exercise 6**
  Let $p(x)=x^5+a_3x^3+a_2x^2+a_1x+a_0$ be the polynomial with real coefficients. Suppose that $x_1=1+\mathrm{i}$ is one root of multiplicity 2. Prove that $a_0=16$.
  - **Solution 6.1**
    $1-\mathrm{i}$ is also a root of multiplication 2. Then $(x-(1+\mathrm{i}))^2(x-(1-\mathrm{i}))^2=x^4-4x^3+8x^2-8x+4 \mid p(x)$. Suppose that $p(x)=(x+r)\left(x^4-4x^3+8x^2-8x+4\right)=x^5+(r-4)x^4+(8-4r)x^3+(8-8r)x^2+(4-8r)x+4r$. Therefore $a_0=4r=16$.